

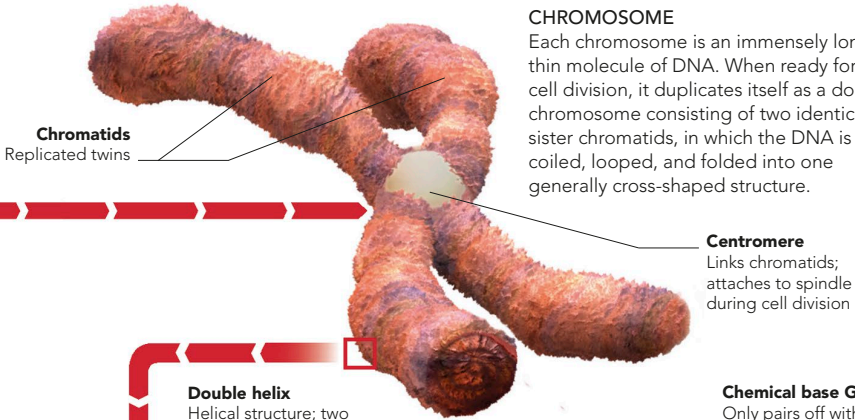
SEQUENCING THE GENOME

By 2003 the Human Genome Project identified all 3 billion base pairs in the full set of human DNA. In 2012 it was realized that a large portion of the DNA instructs for building RNA rather than proteins. A major technique used in DNA sequencing is gel electrophoresis. DNA is extracted from cells, purified, and broken into smaller fragments of known length by chemicals known as restriction enzymes. The DNA fragments are separated out and stained with dye, showing up as dark stripes, like bar codes (see right). Computers can read these bar codes and reveal the sequences of base pairs.



CHROMOSOME

Each chromosome is an immensely long, thin molecule of DNA. When ready for cell division, it duplicates itself as a double-chromosome consisting of two identical sister chromatids, in which the DNA is coiled, looped, and folded into one generally cross-shaped structure.

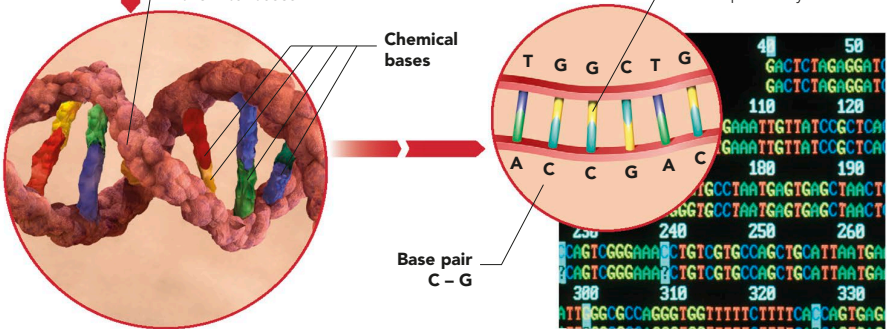


Double helix

Helical structure; two backbones joined by chemical bases

Chemical base G

Only pairs off with chemical base C; A pairs only with T



GENETIC CODE

DNA consists of two spiral backbones joined by cross-rungs, which are pairs of chemical bases. The bases are adenine (A), thymine (T), guanine (G), and cytosine (C).

GENETIC SEQUENCE

The order of base pairs on DNA represents the coded genetic information. Using chemicals to identify the bases, DNA sequencing machines can show the data on screen as lists of letters.

